Program / Procedure

Topology: PC0 (192.168.1.2/24) – R1 – serial link 10.0.0.0/30 – R2 – PC1 (192.168.2.2/24).

Router R1:

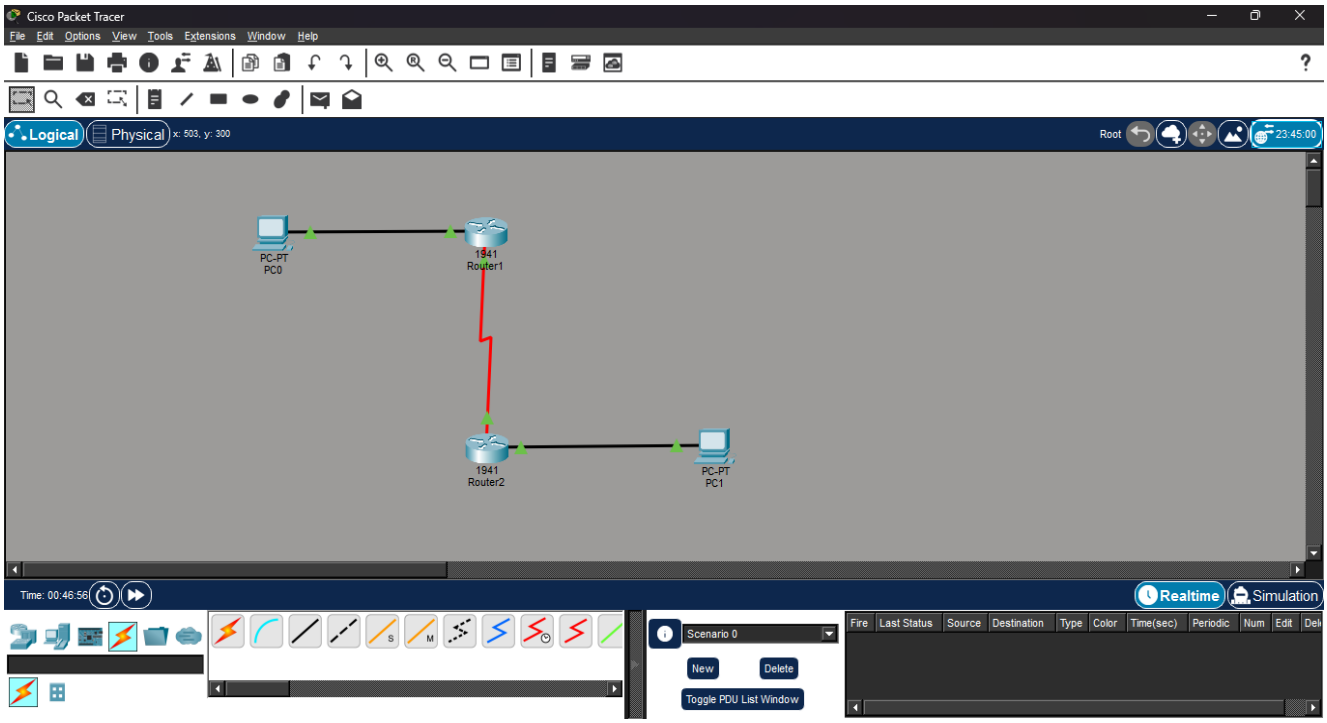
```
Router> enable
Router# configure terminal
Router(config)# hostname R1
R1(config)# interface gigabitEthernet 0/0
R1(config-if)# ip address 192.168.1.1 255.255.255.0
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)# interface serial 0/0/0
R1(config-if)# ip address 10.0.0.1 255.255.255.252
R1(config-if)# clock rate 64000
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)# ip route 192.168.2.0 255.255.255.0 10.0.0.2
R1(config)# end
R1# write memory
```

Router R2:

```
Router(config)# hostname R2
R2(config)# interface gigabitEthernet 0/0
R2(config-if)# ip address 192.168.2.1 255.255.255.0
R2(config-if)# no shutdown
R2(config-if)# exit
R2(config)# interface serial 0/0/0
R2(config-if)# ip address 10.0.0.2 255.255.255.252
R2(config-if)# no shutdown
R2(config-if)# exit
R2(config)# ip route 192.168.1.0 255.255.255.0 10.0.0.1
```

Verification: set the PC gateways to 192.168.1.1 and 192.168.2.1, then run:

```
R1# show ip route
PC0> ping 192.168.2.2
PC0> tracert 192.168.2.2
```



PC0

Physical Config Desktop Programming Attributes

Command Prompt

```
Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time=1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.4

Pinging 192.168.1.4 with 32 bytes of data:

Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time=1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.1.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Top

PC0

Physical Config Desktop Programming Attributes

Command Prompt

```
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.2.2: bytes=32 time=1ms TTL=126
Reply from 192.168.2.2: bytes=32 time=1ms TTL=126
Reply from 192.168.2.2: bytes=32 time=1ms TTL=126
Reply from 192.168.2.2: bytes=32 time=3ms TTL=126

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 3ms, Average = 1ms

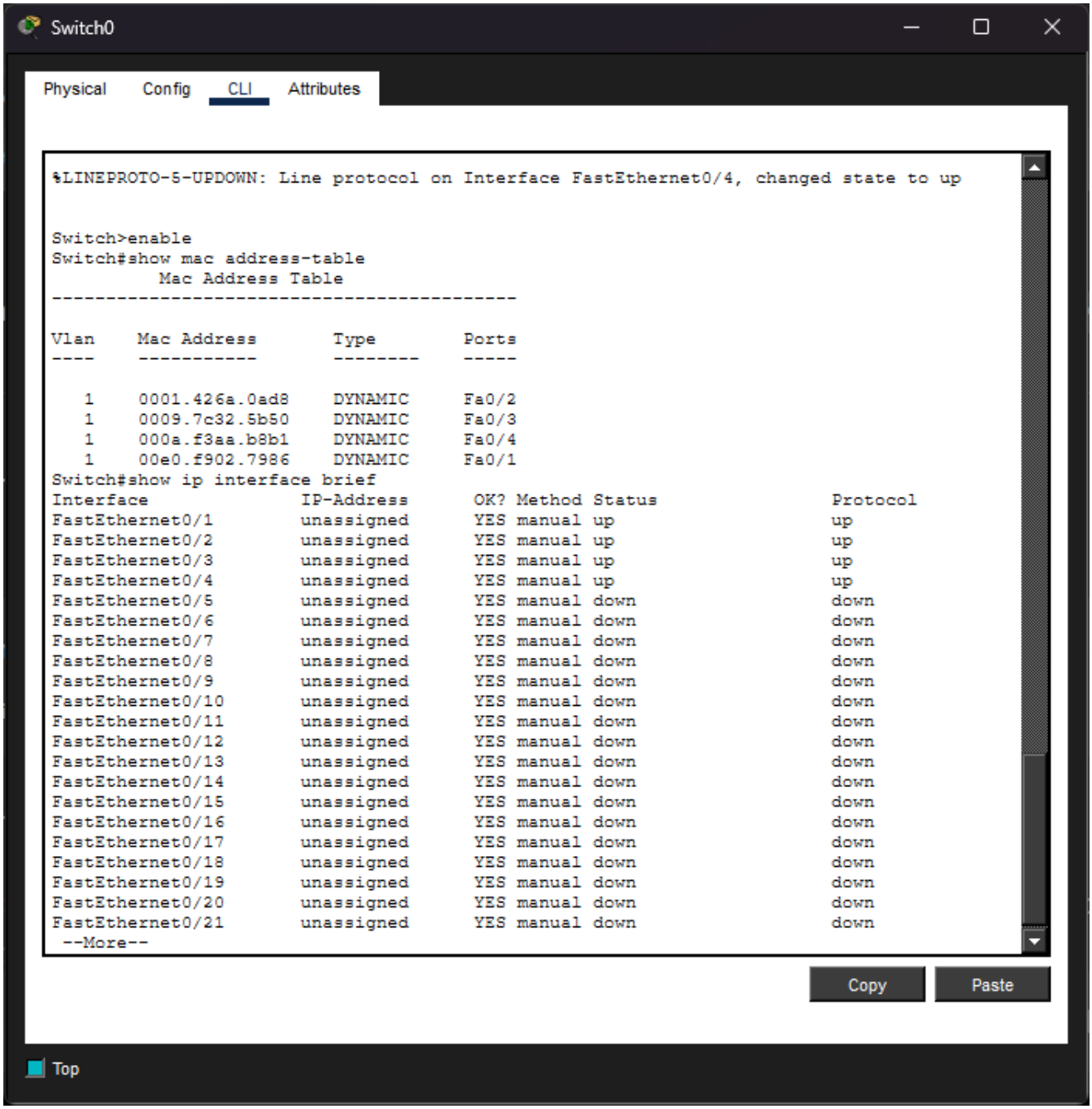
C:\>tracert 192.168.2.2

Tracing route to 192.168.2.2 over a maximum of 30 hops:

  1    0 ms      0 ms      0 ms      192.168.1.1
  2    1 ms      2 ms      1 ms      10.0.0.2
  3    1 ms      1 ms      3 ms      192.168.2.2

Trace complete.

C:\>
```


Switch0
Physical
Config
CLI
Attributes
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up
Switch>enable
Switch#show mac address-table
Mac Address Table
Vlan Mac Address Type Ports
1 0001.426a.0ad8 DYNAMIC Fa0/2
1 0009.7c32.5b50 DYNAMIC Fa0/3
1 000a.f3aa.b8b1 DYNAMIC Fa0/4
1 00e0.f902.7986 DYNAMIC Fa0/1
Switch#show ip interface brief
Interface IP-Address OK? Method Status Protocol
FastEthernet0/1 unassigned YES manual up up
FastEthernet0/2 unassigned YES manual up up
FastEthernet0/3 unassigned YES manual up up
FastEthernet0/4 unassigned YES manual up up
FastEthernet0/5 unassigned YES manual down down
FastEthernet0/6 unassigned YES manual down down
FastEthernet0/7 unassigned YES manual down down
FastEthernet0/8 unassigned YES manual down down
FastEthernet0/9 unassigned YES manual down down
FastEthernet0/10 unassigned YES manual down down
FastEthernet0/11 unassigned YES manual down down
FastEthernet0/12 unassigned YES manual down down
FastEthernet0/13 unassigned YES manual down down
FastEthernet0/14 unassigned YES manual down down
FastEthernet0/15 unassigned YES manual down down
FastEthernet0/16 unassigned YES manual down down
FastEthernet0/17 unassigned YES manual down down
FastEthernet0/18 unassigned YES manual down down
FastEthernet0/19 unassigned YES manual down down
FastEthernet0/20 unassigned YES manual down down
FastEthernet0/21 unassigned YES manual down down
--More--
Copy
Paste
Top

Router1

Physical Config CLI Attributes

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, Serial0/1/0
L       10.0.0.1/32 is directly connected, Serial0/1/0
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/1
L       192.168.1.1/32 is directly connected, GigabitEthernet0/1
O    192.168.2.0/24 [110/65] via 10.0.0.2, 00:00:50, Serial0/1/0
```

Router1

Physical Config CLI Attributes

```
R1#show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
192.168.2.1       0   FULL/  -        00:00:36    10.0.0.2        Serial0/1/0
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, Serial0/1/0
L       10.0.0.1/32 is directly connected, Serial0/1/0
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/1
L       192.168.1.1/32 is directly connected, GigabitEthernet0/1
O    192.168.2.0/24 [110/65] via 10.0.0.2, 00:08:19, Serial0/1/0

R1#
```

PC0

Physical Config Desktop Programming Attributes

Command Prompt X

```
C:\>ping 192.168.2.2

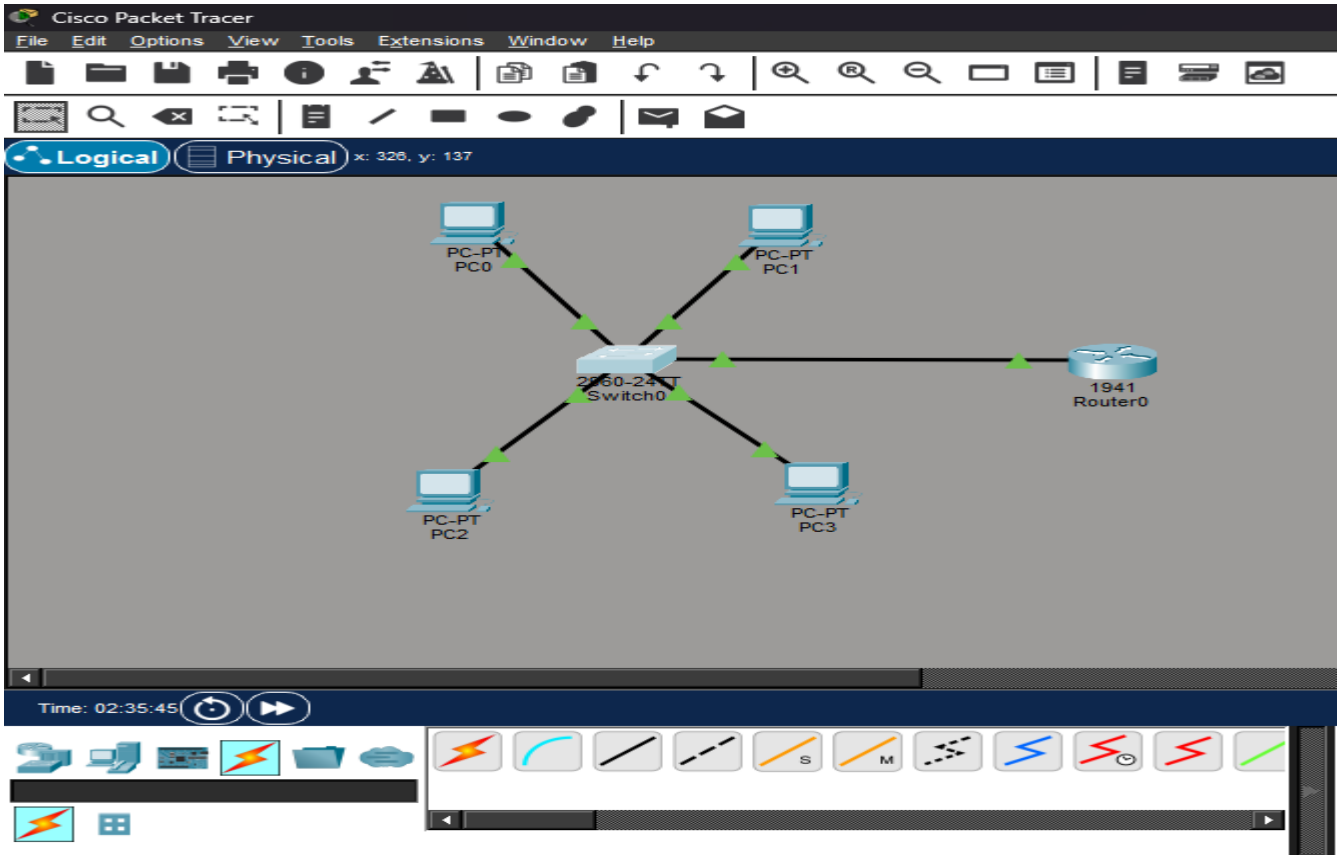
Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.2.2: bytes=32 time=10ms TTL=126
Reply from 192.168.2.2: bytes=32 time=1ms TTL=126
Reply from 192.168.2.2: bytes=32 time=2ms TTL=126
Reply from 192.168.2.2: bytes=32 time=3ms TTL=126

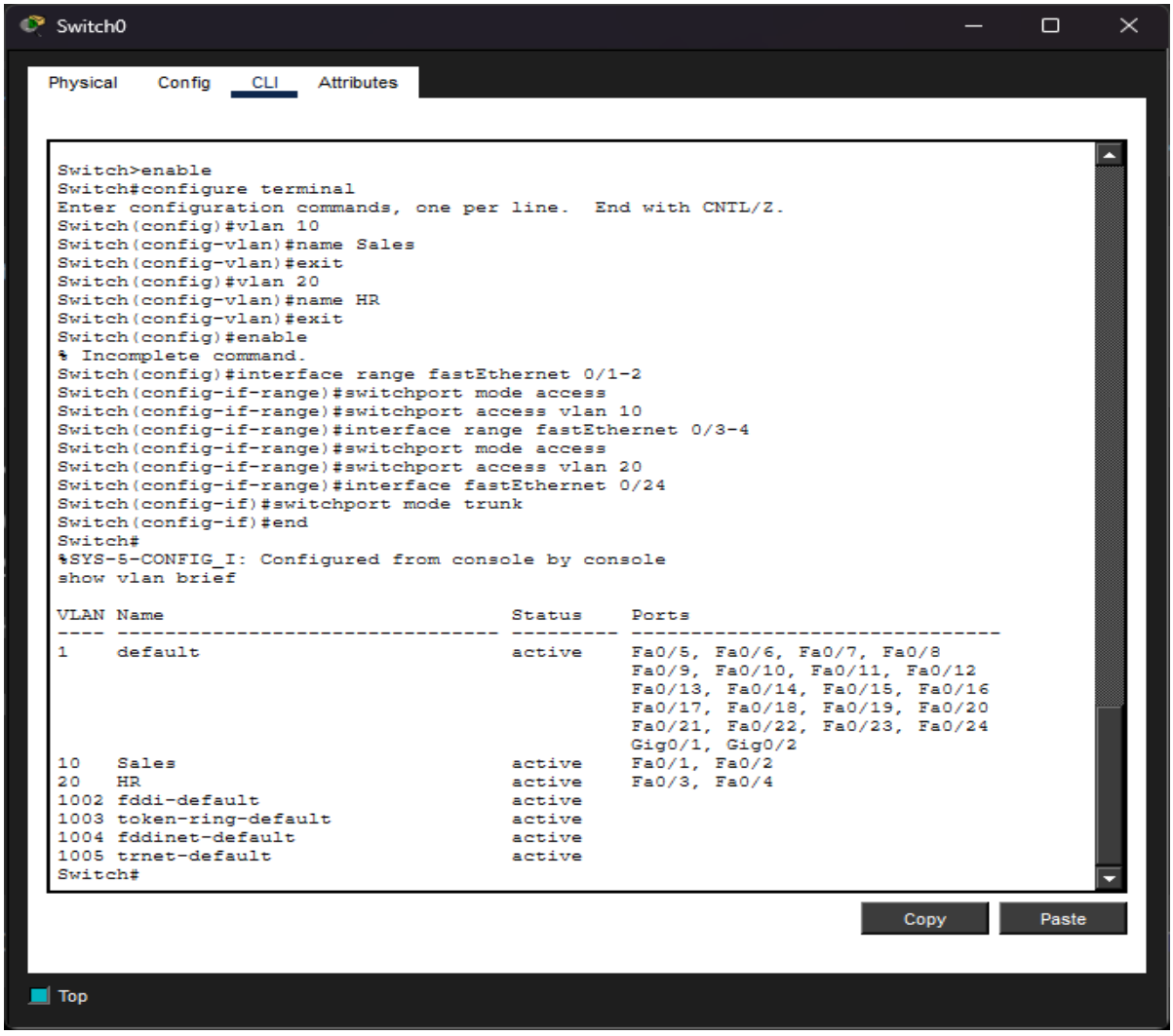
Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 10ms, Average = 4ms

C:\>
```

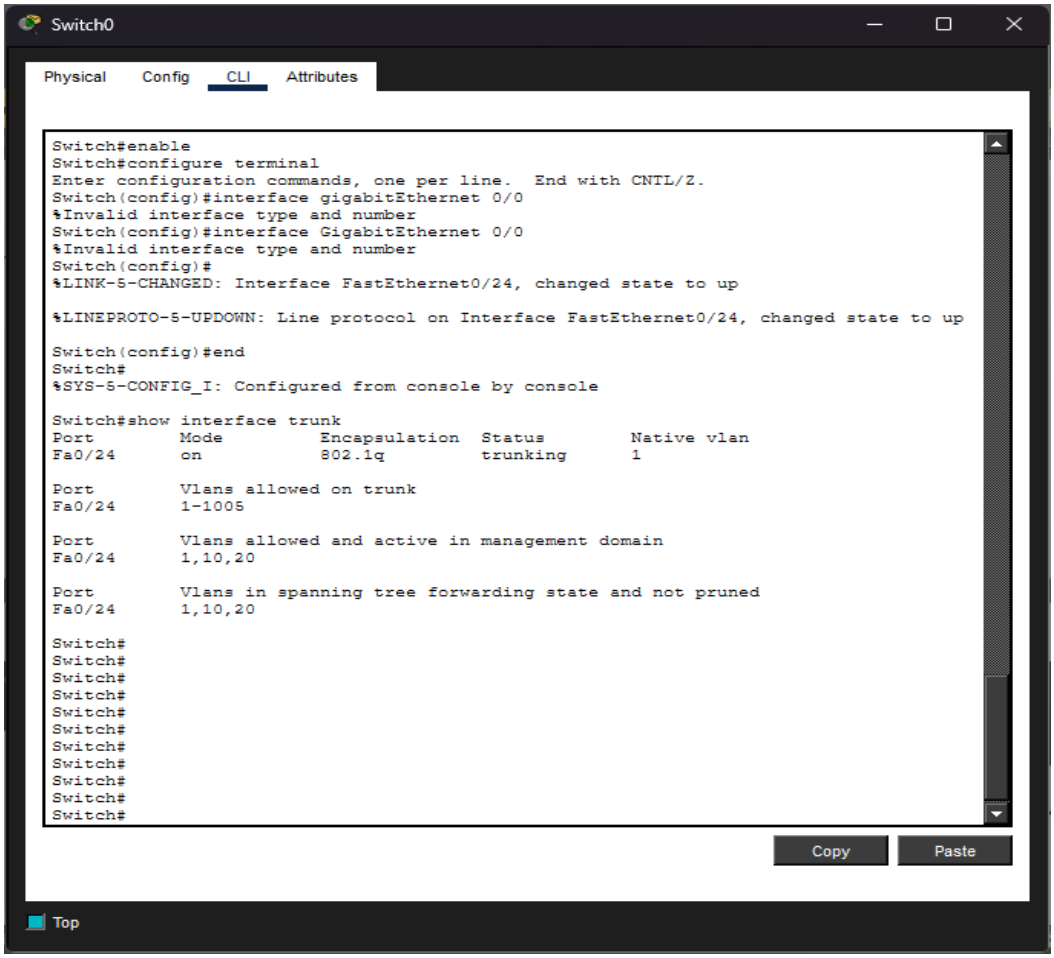
1. **Complete topology** — 4 PCs + Switch + Router



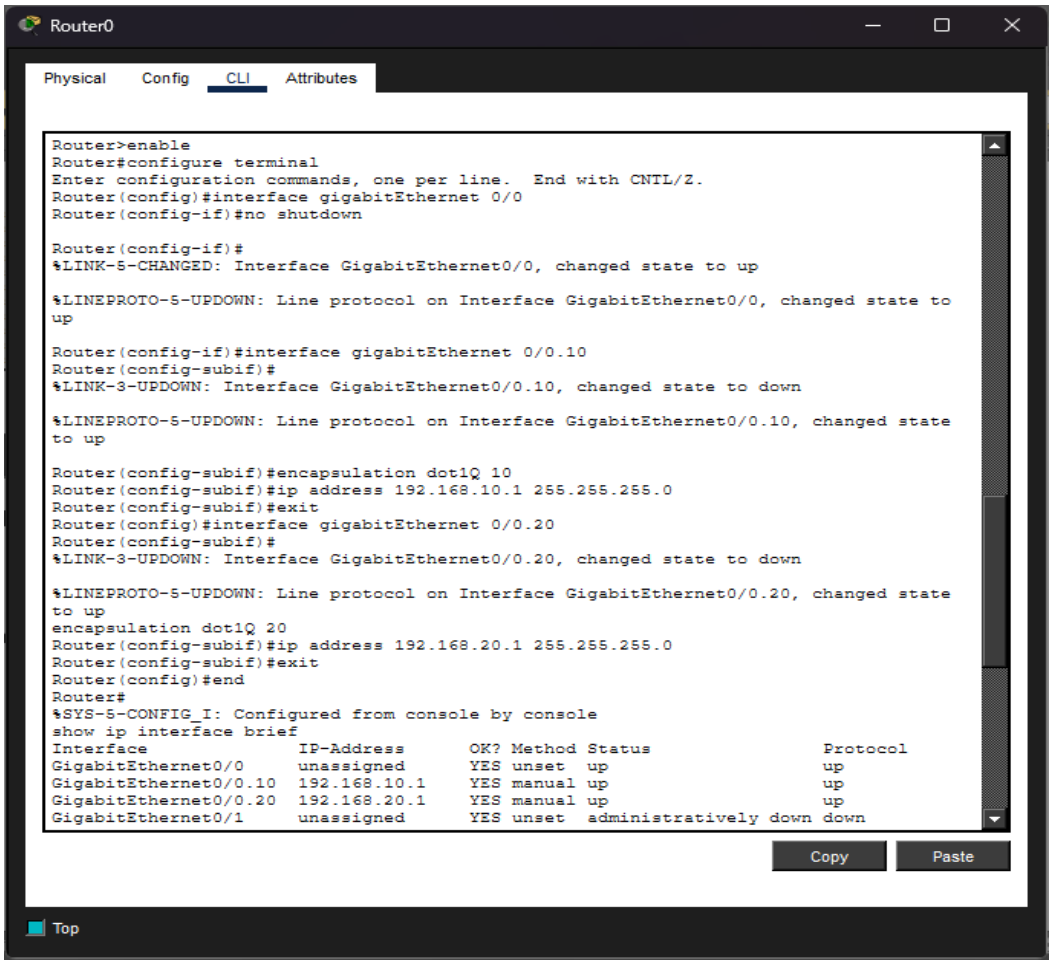
2. **show vlan brief** — VLAN 10/20 aur ports visible



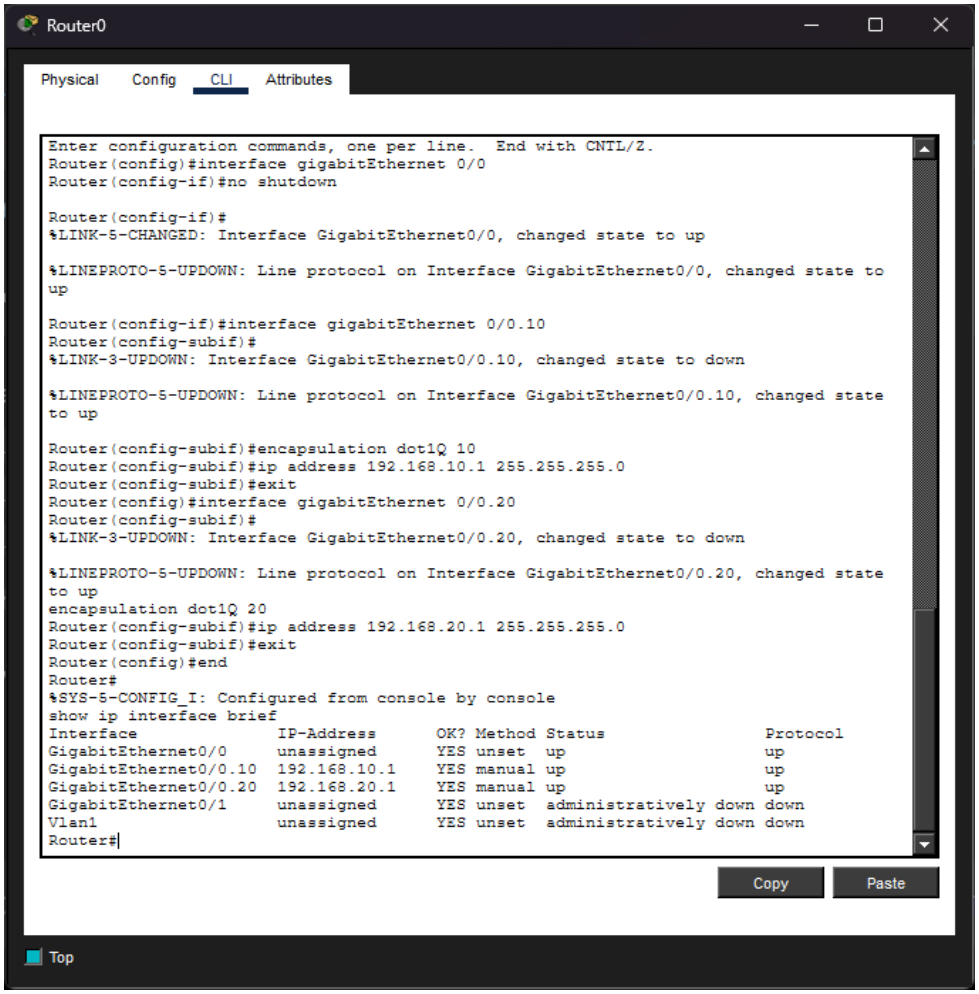
3. **show interfaces trunk** — trunk port visible



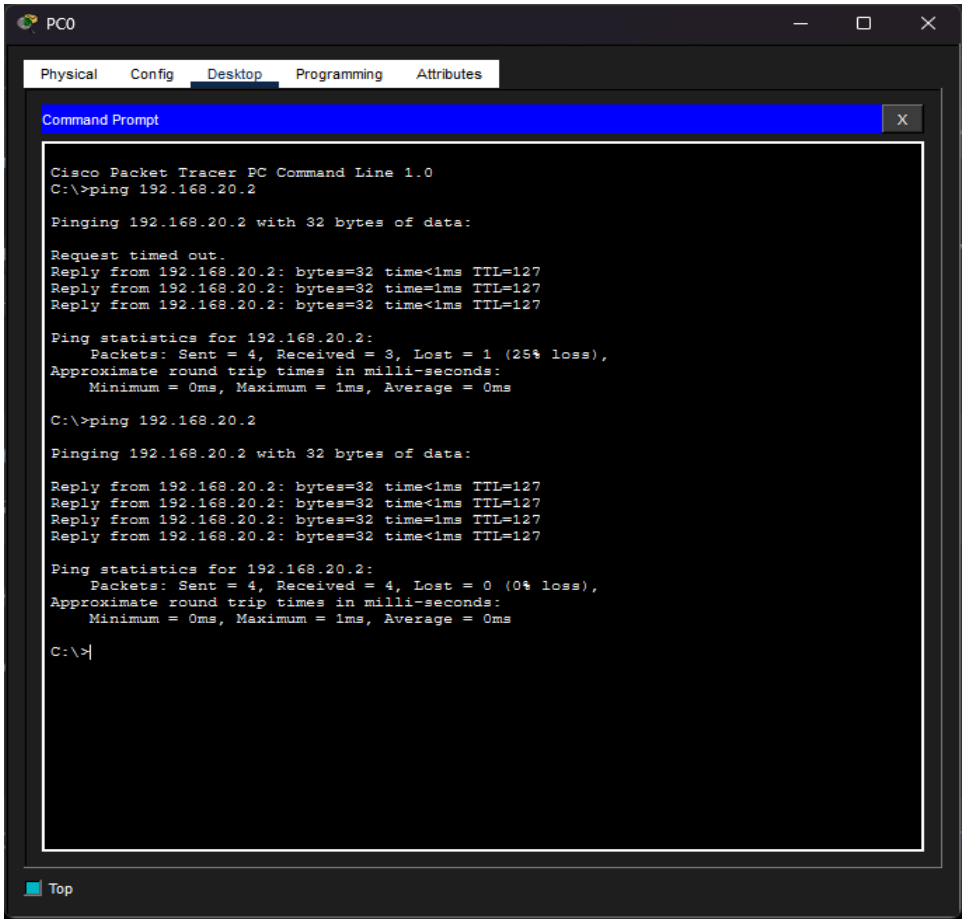
4. **Router sub-interface configuration** — VLAN 10 & VLAN 20



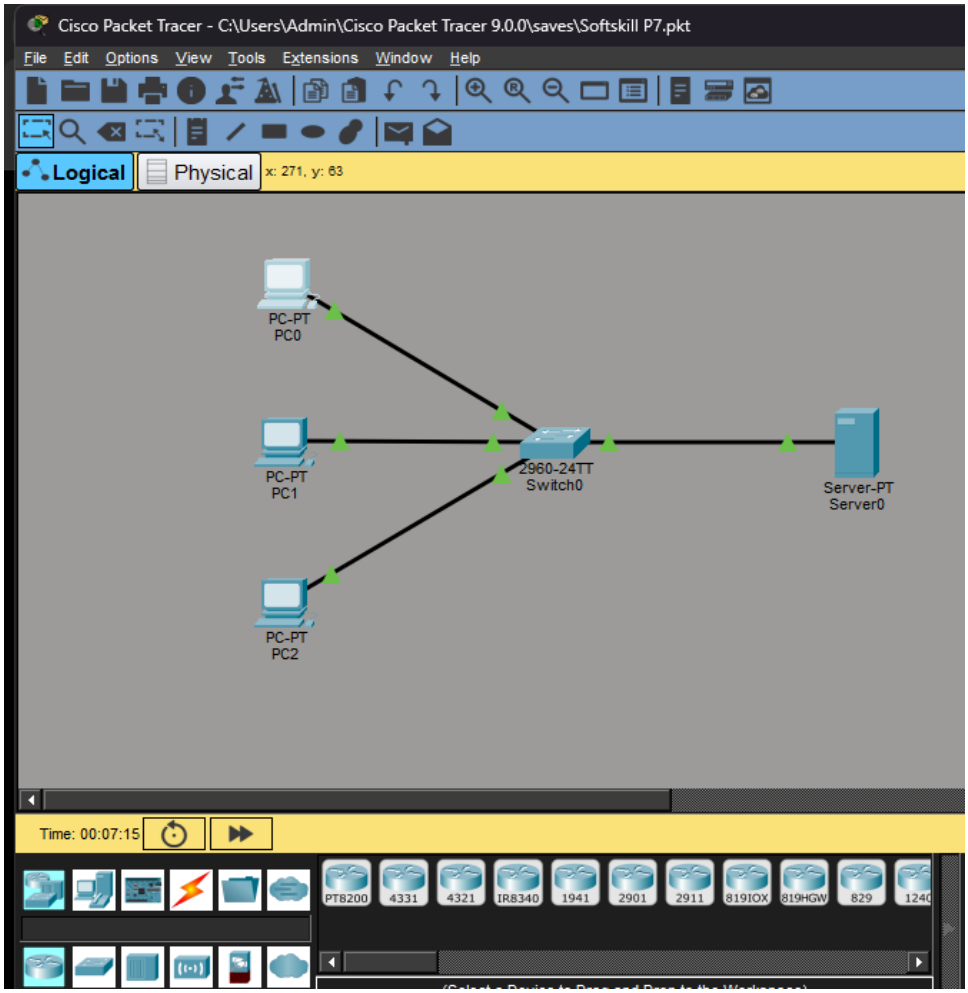
5. **show ip interface brief** — router interfaces/subinterfaces up



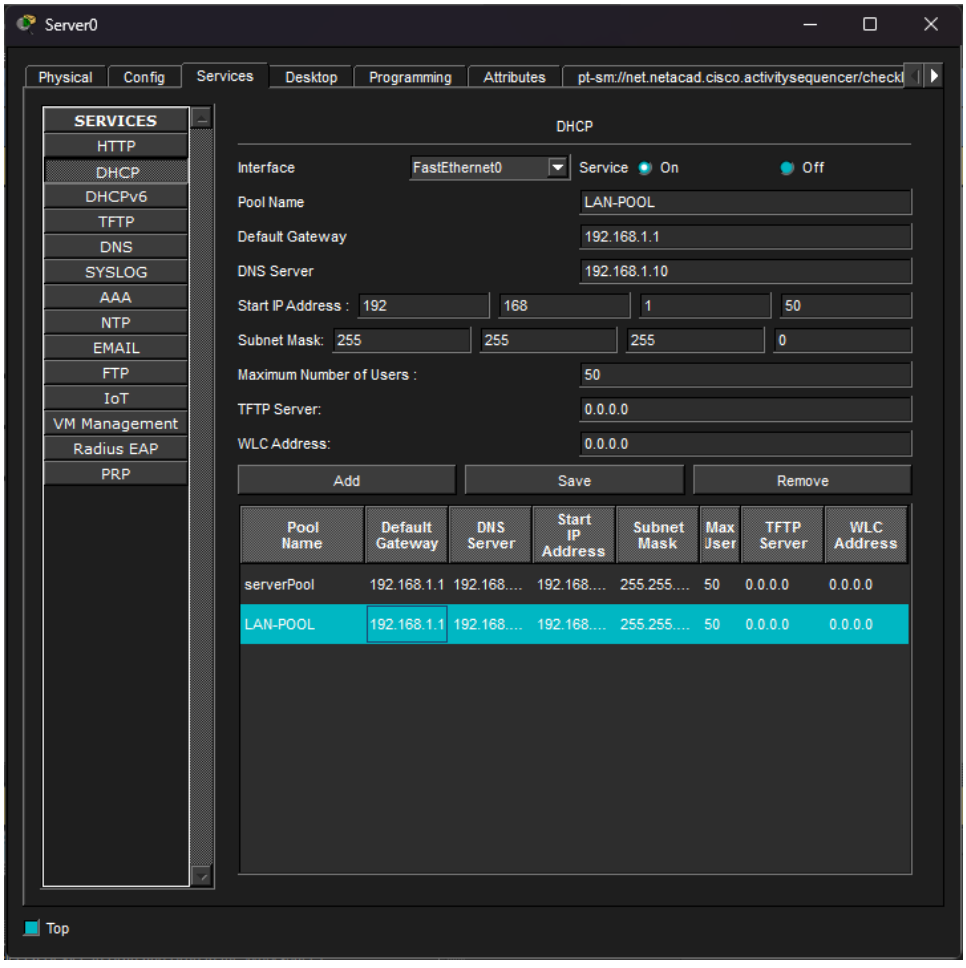
6. **Final inter-VLAN ping** — PC1 → PC3 successful



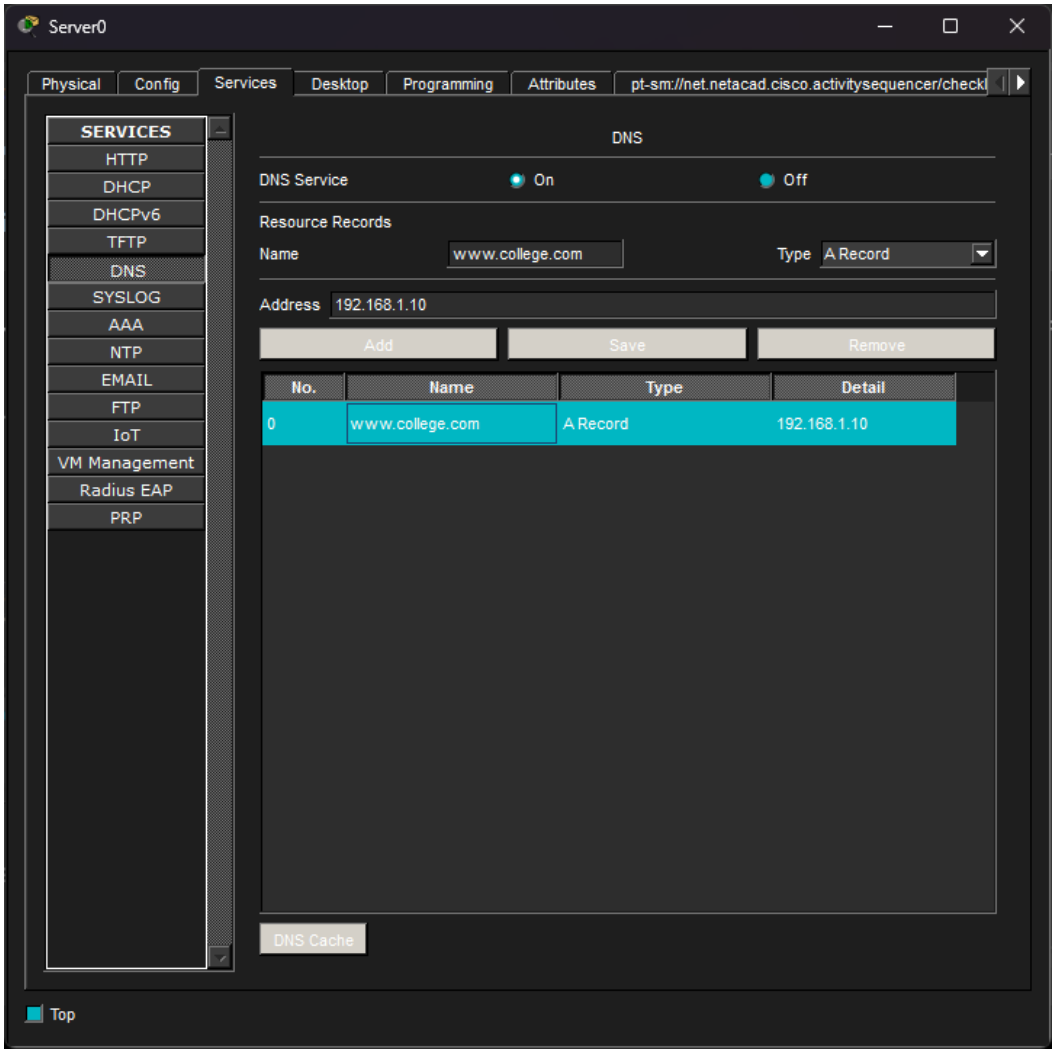
1. Complete topology



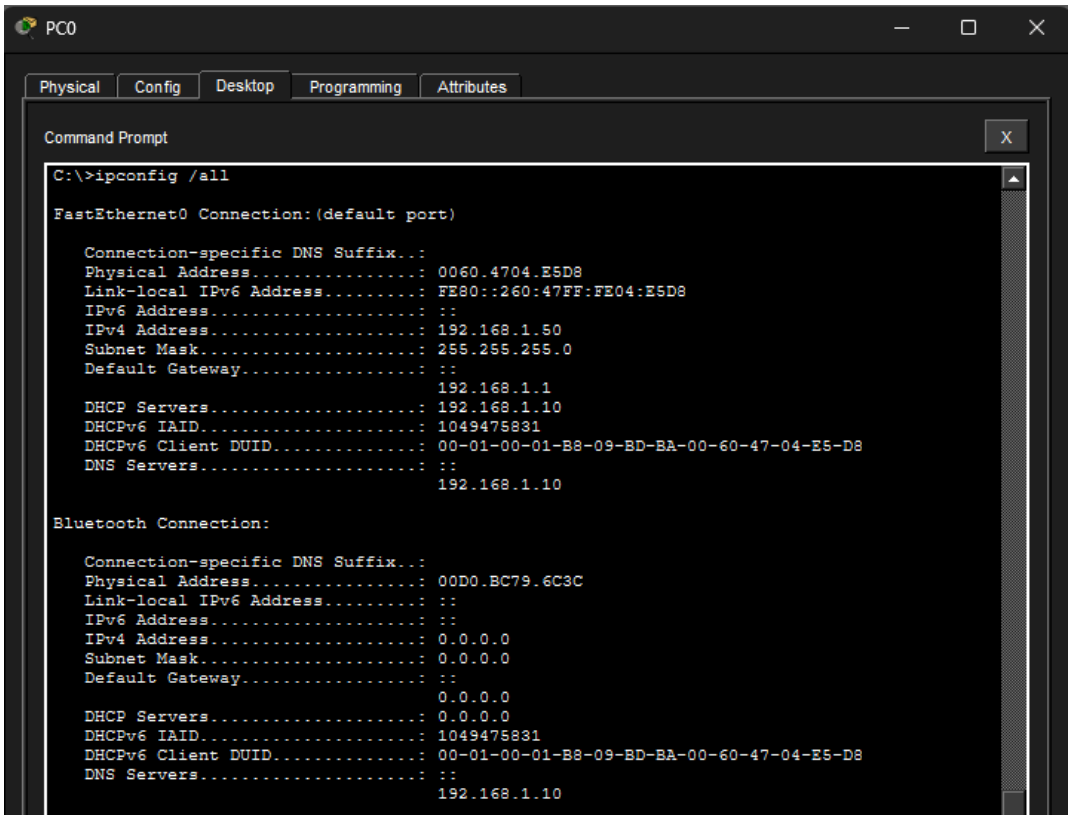
2. Server DHCP configuration



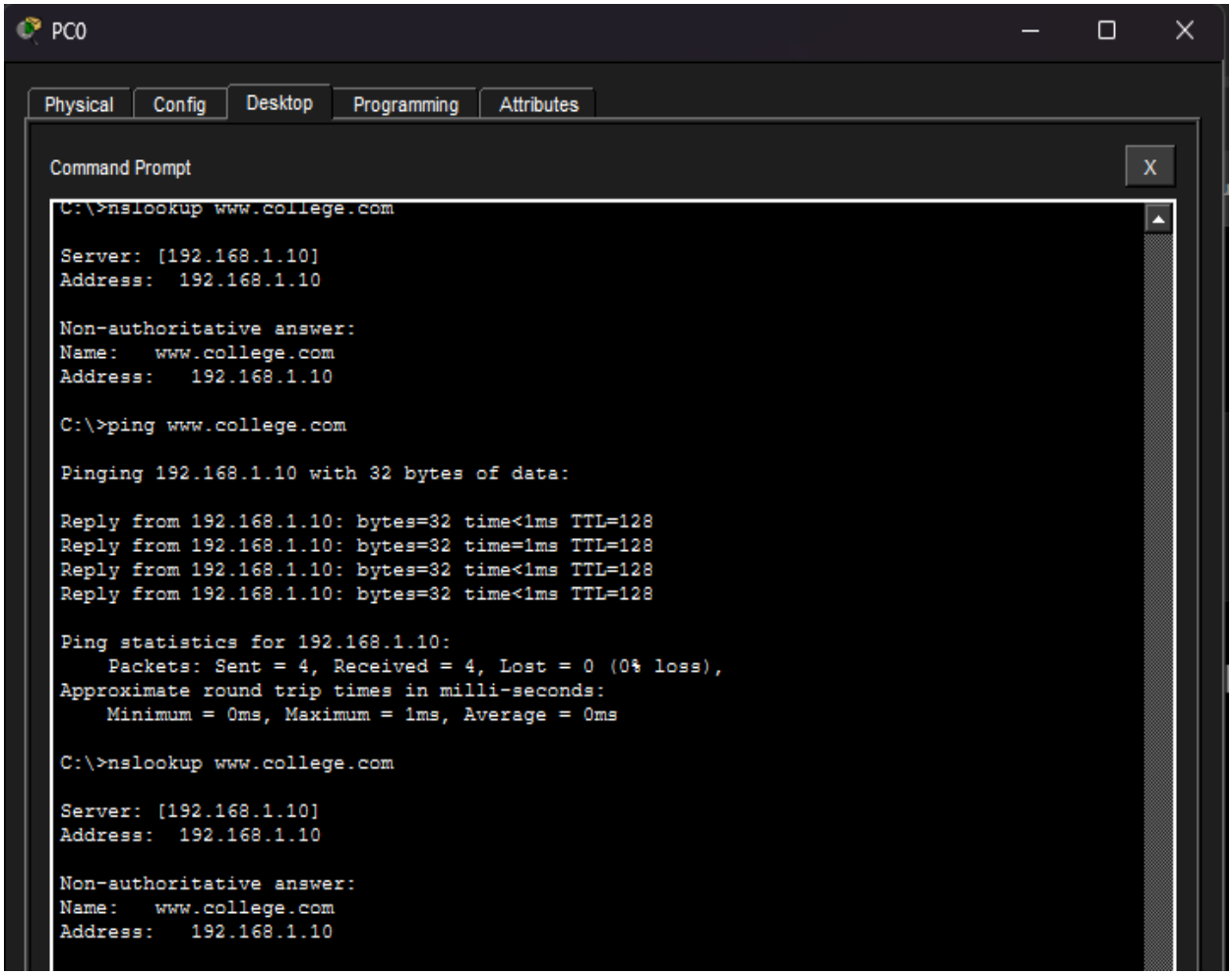
3. Server DNS configuration



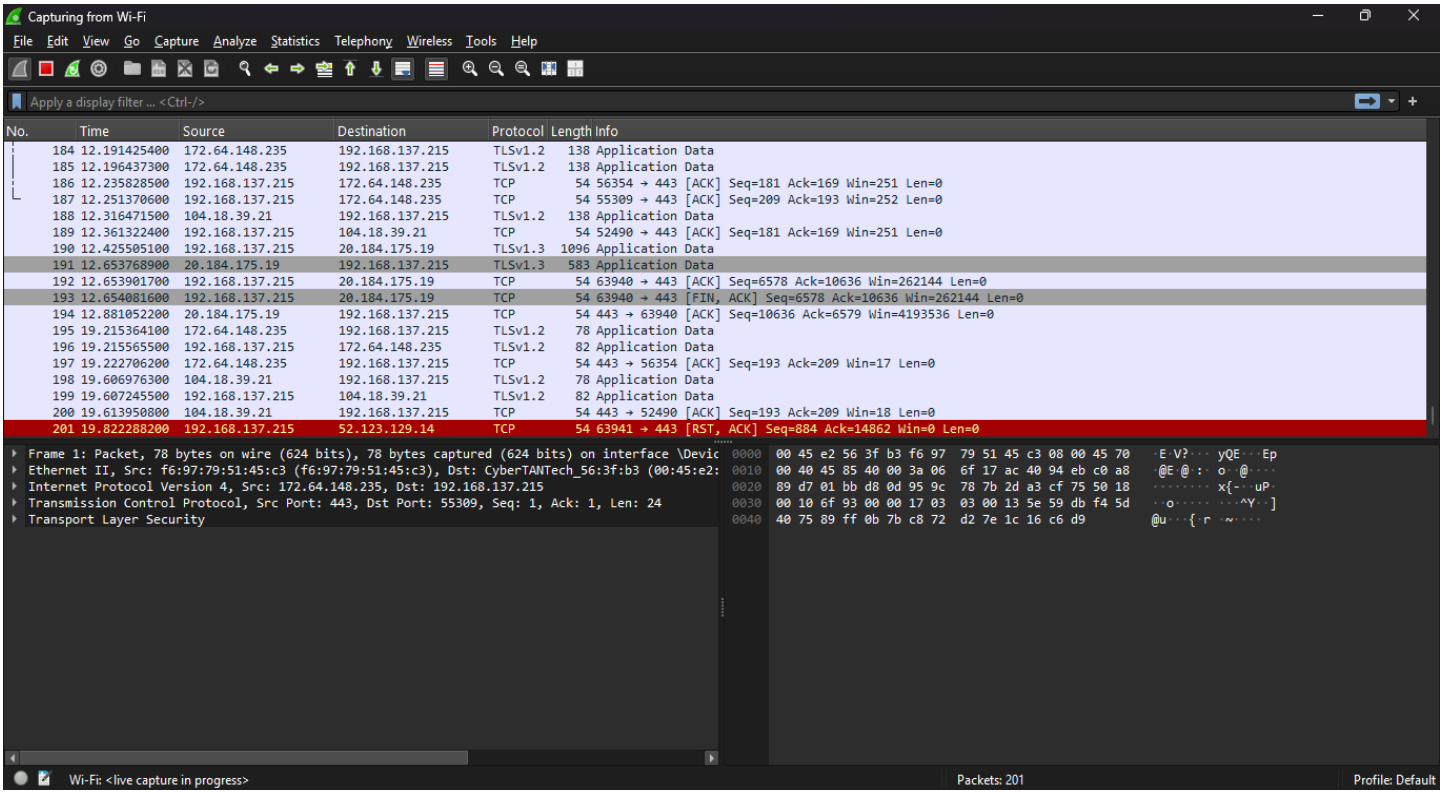
4. PC ka ipconfig /all showing DHCP address



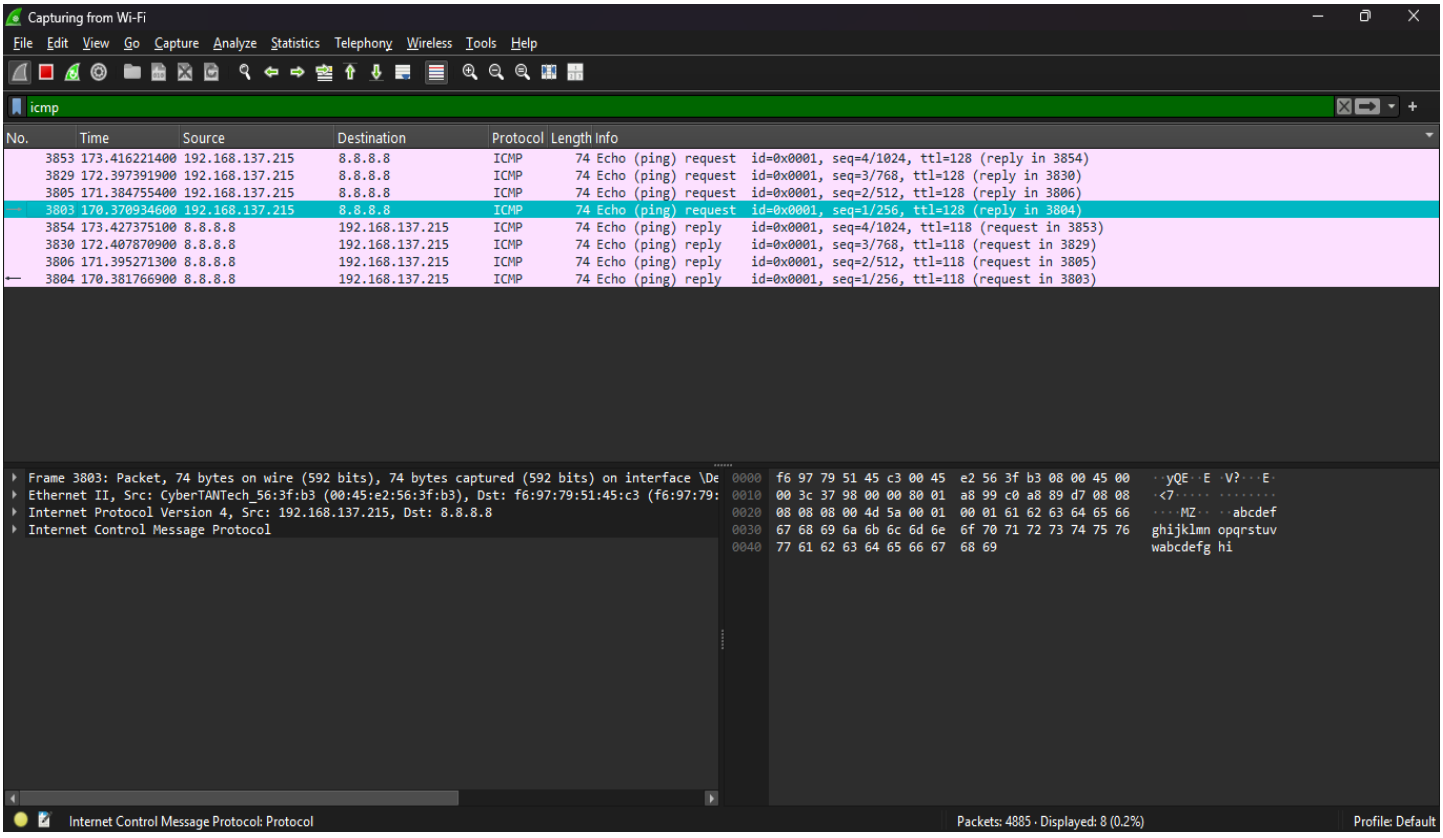
5. ping www.college.com + nslookup www.college.com



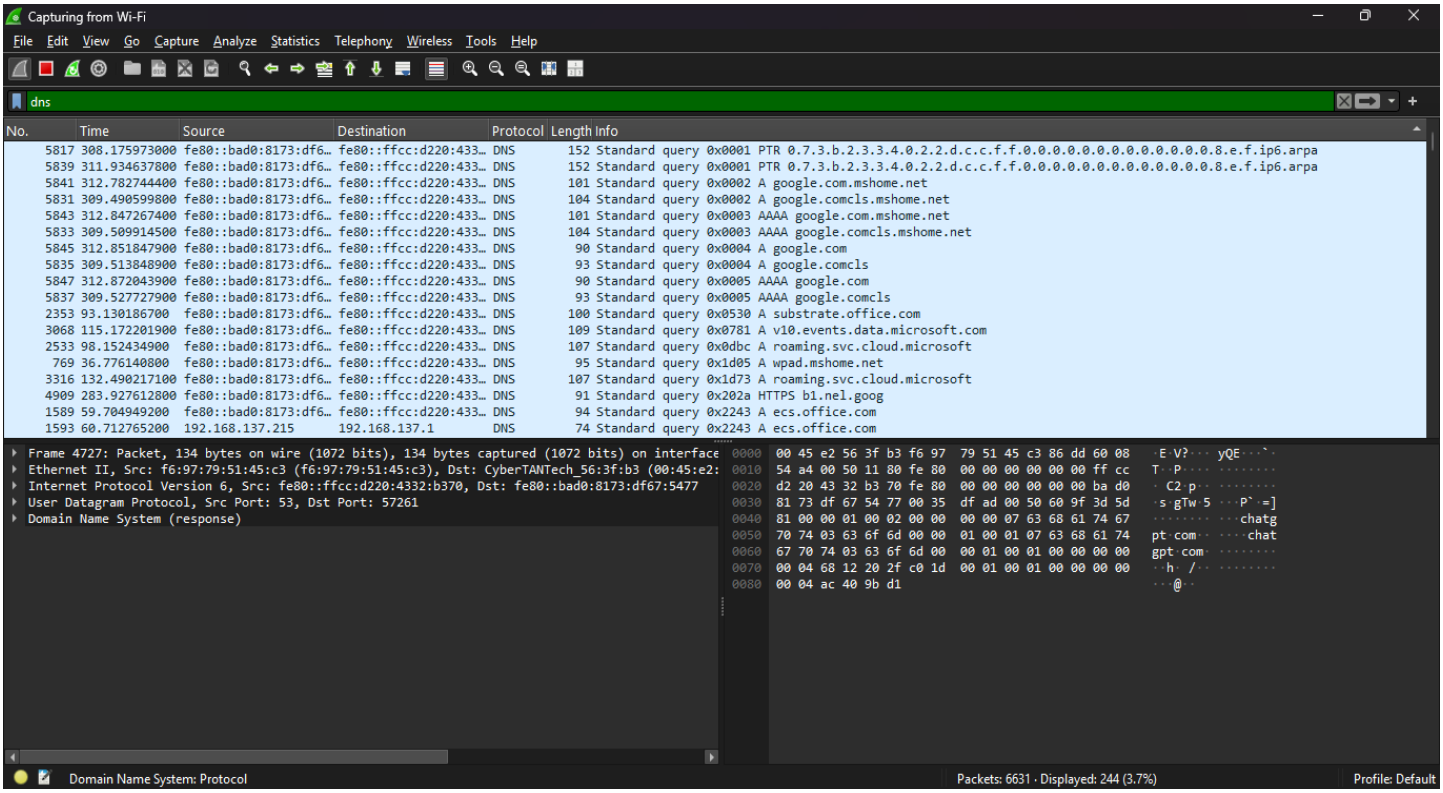
1. Wireshark live capture



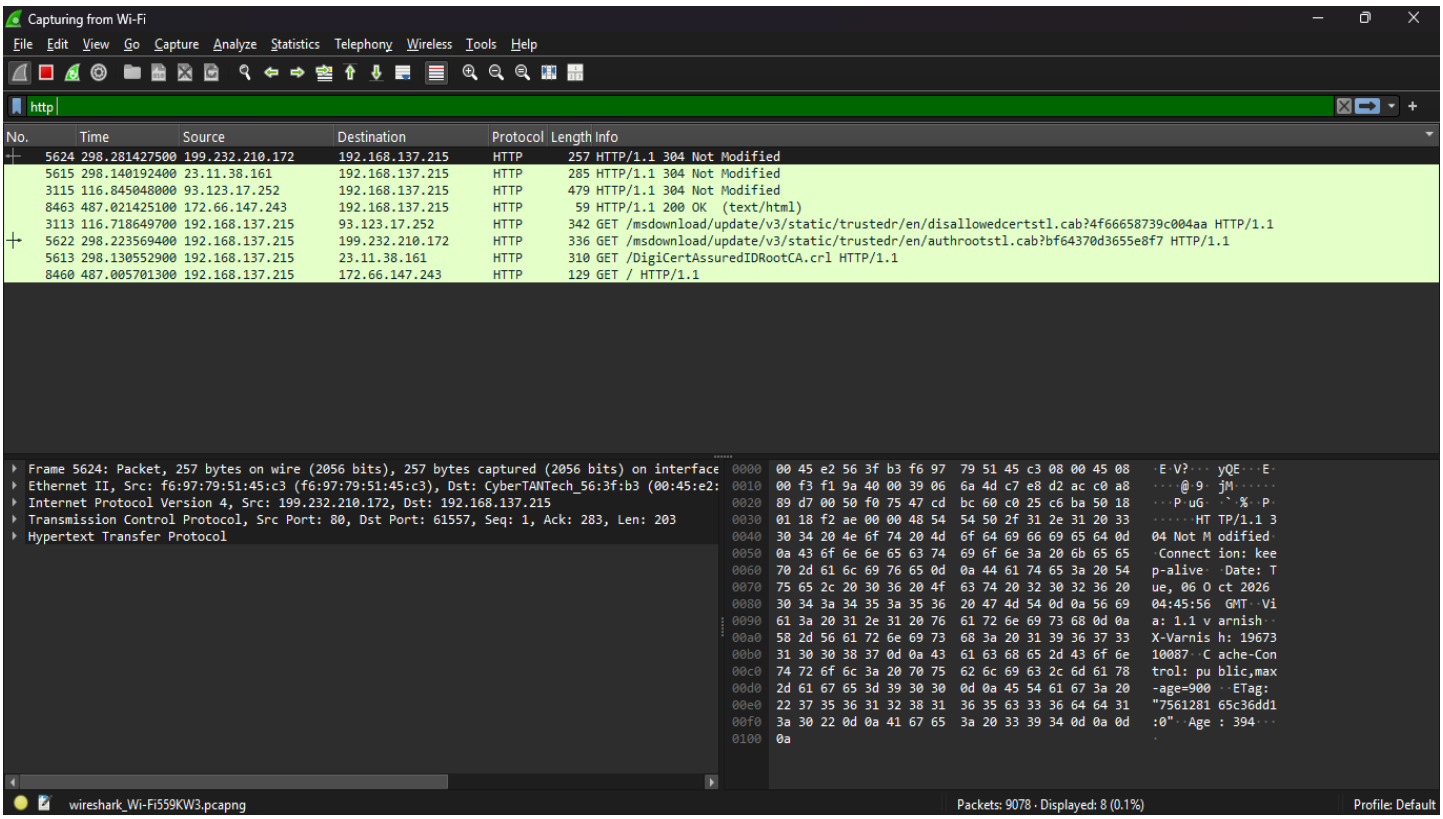
2. ICMP packet (ping)



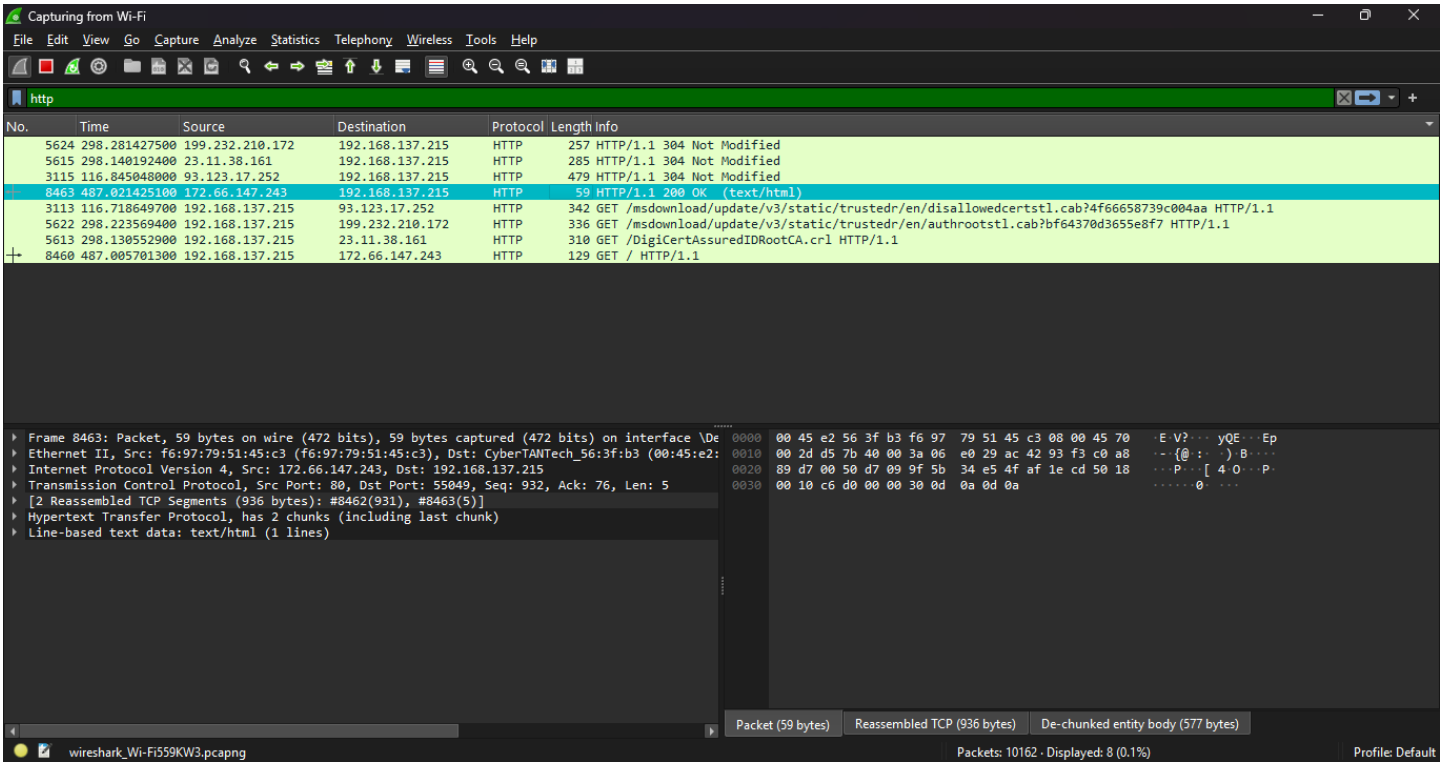
3. DNS packet (nslookup)

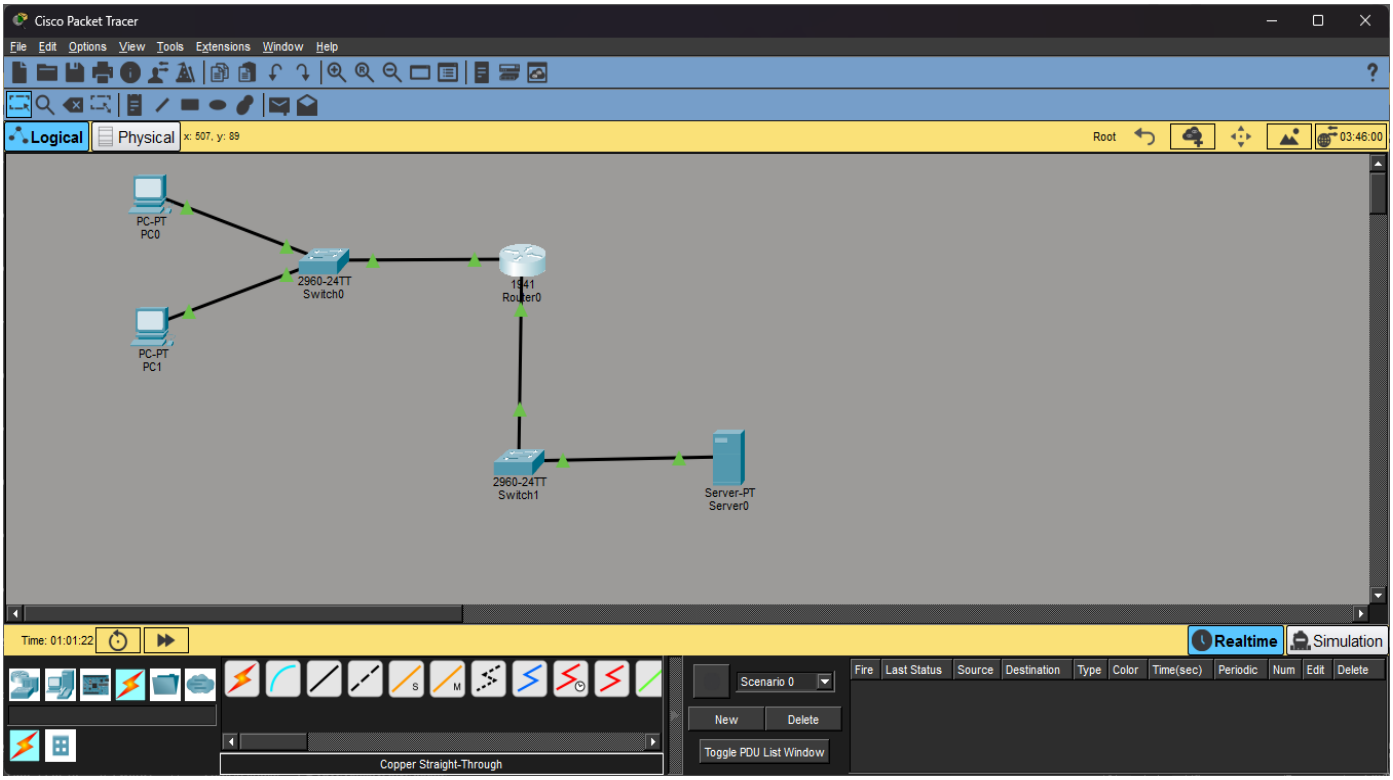


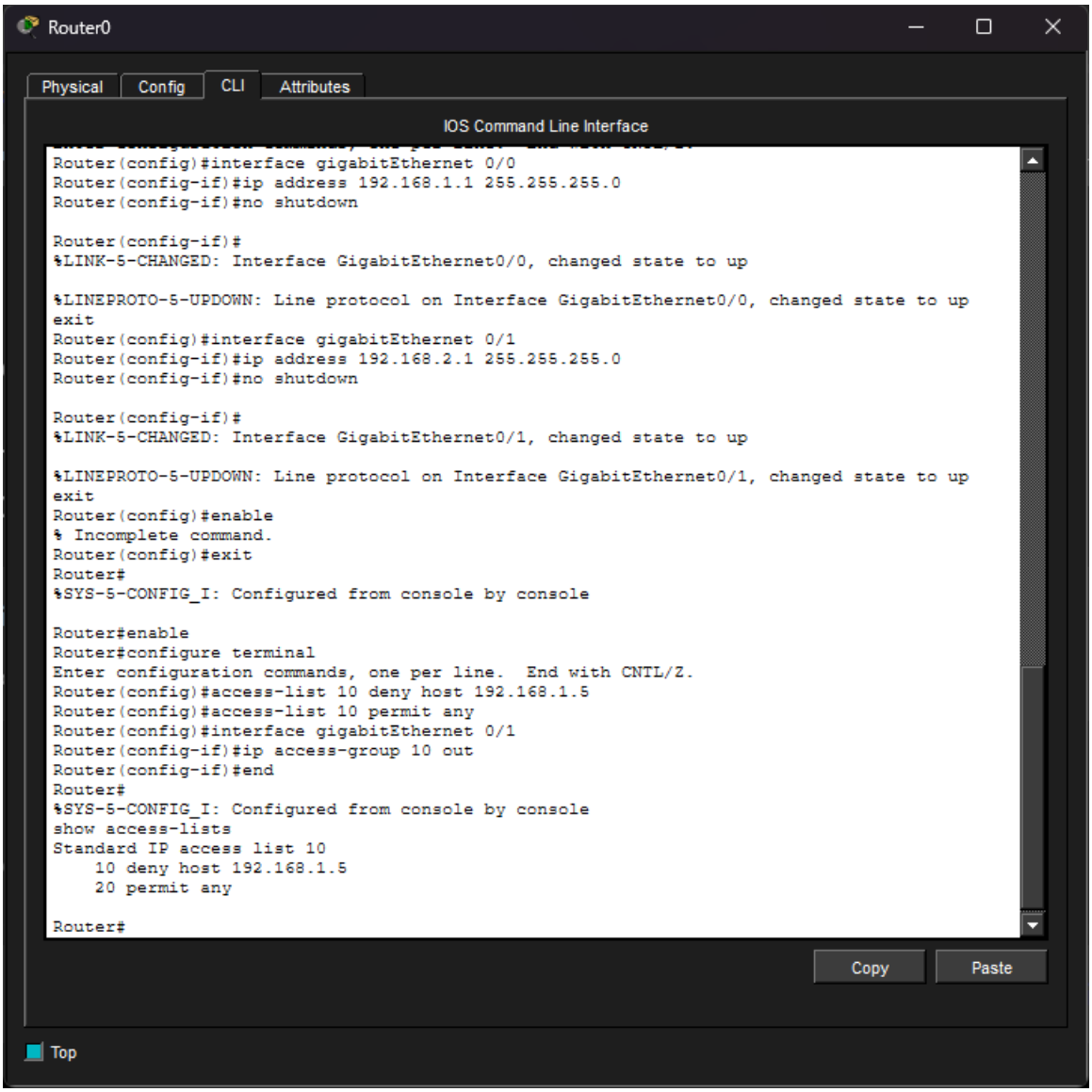
4. TCP/HTTP packet

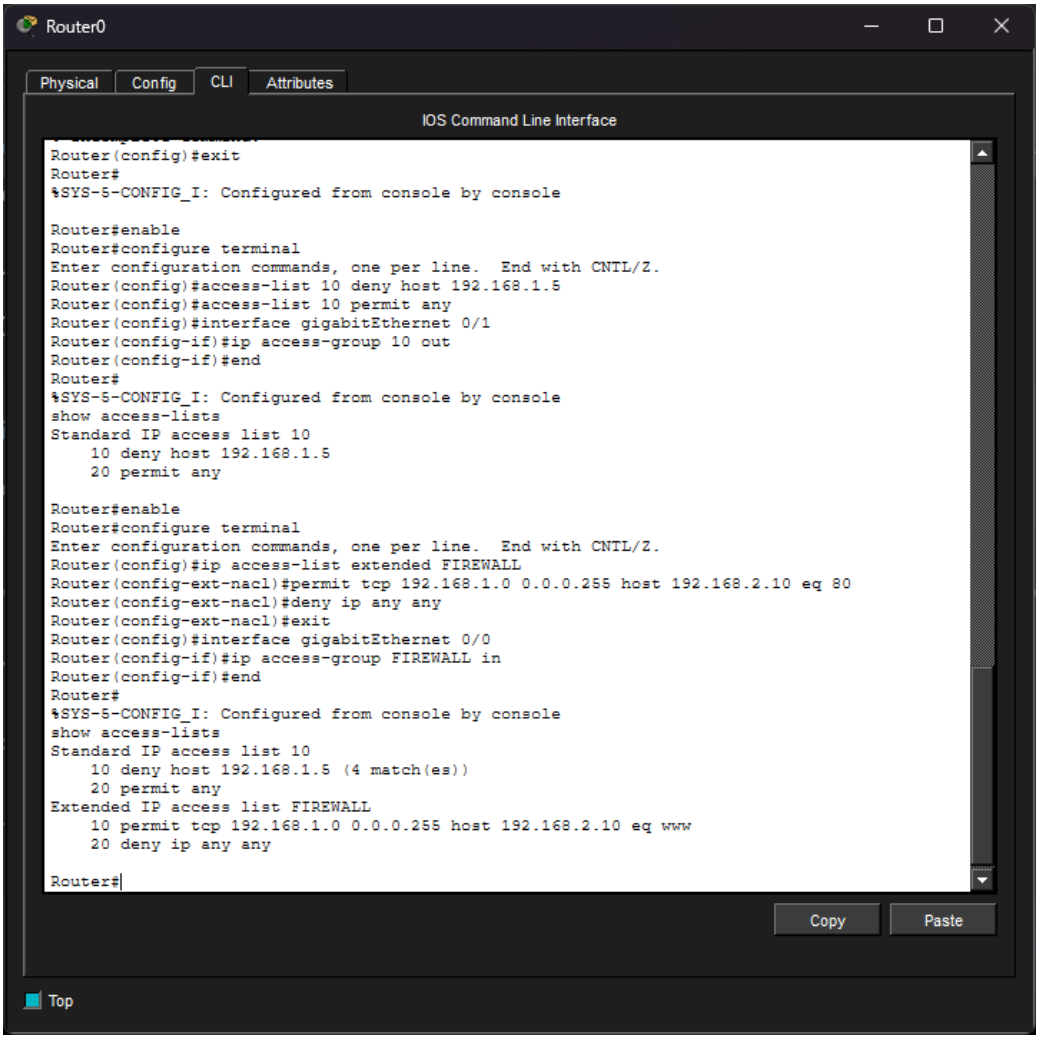


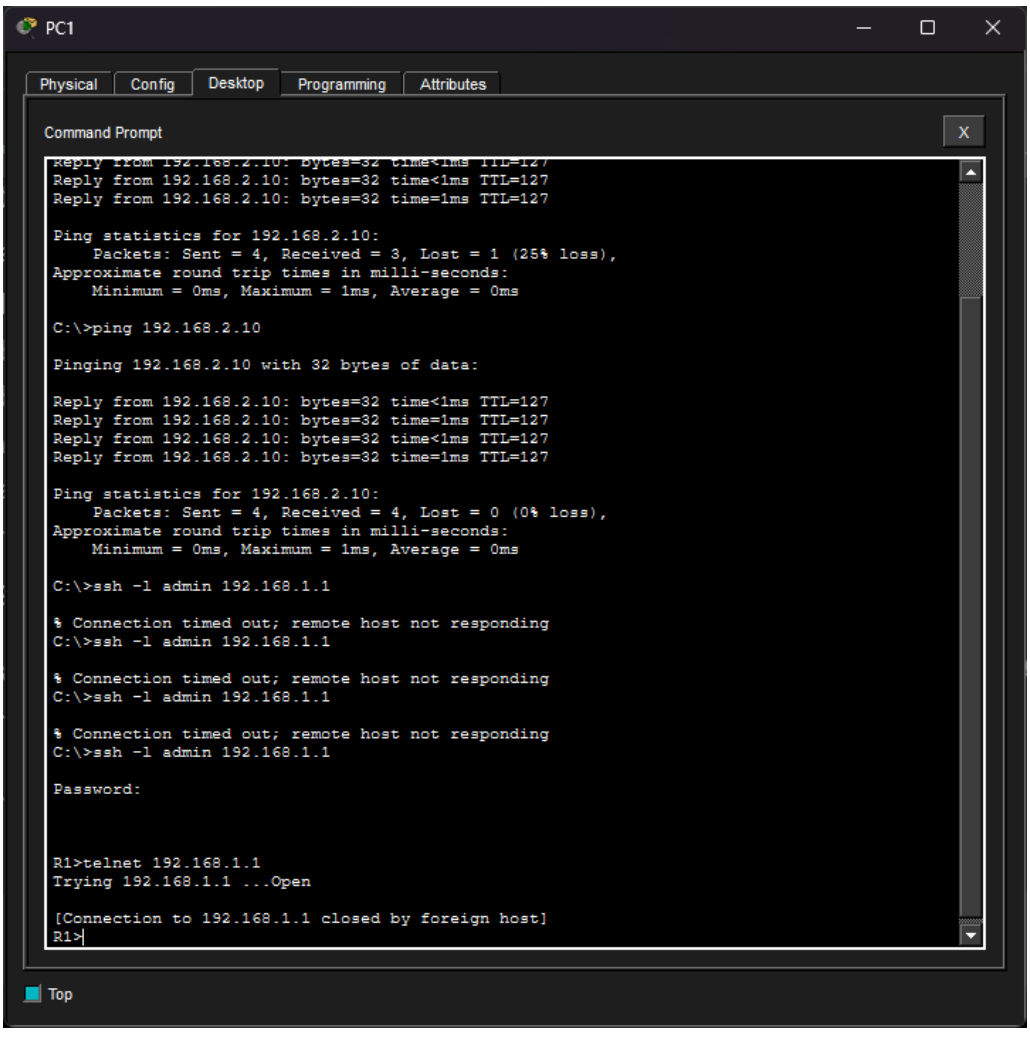
5. Packet Details pane with expanded layers



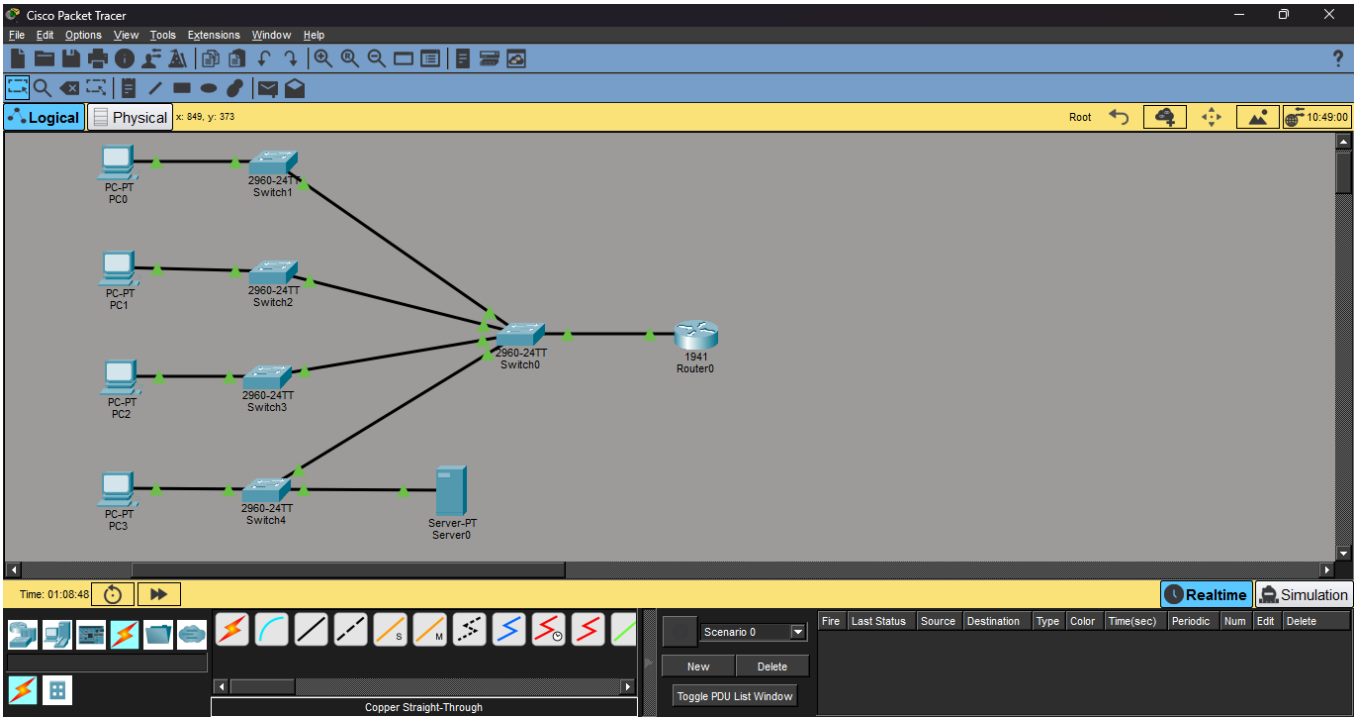

Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical
Physical
Root
PC-PT PC0
2960-24TT Switch0
1941 Router0
PC-PT PC1
2960-24TT Switch1
Server-PT Server0
Realtime
Simulation
Copper Straight-Through

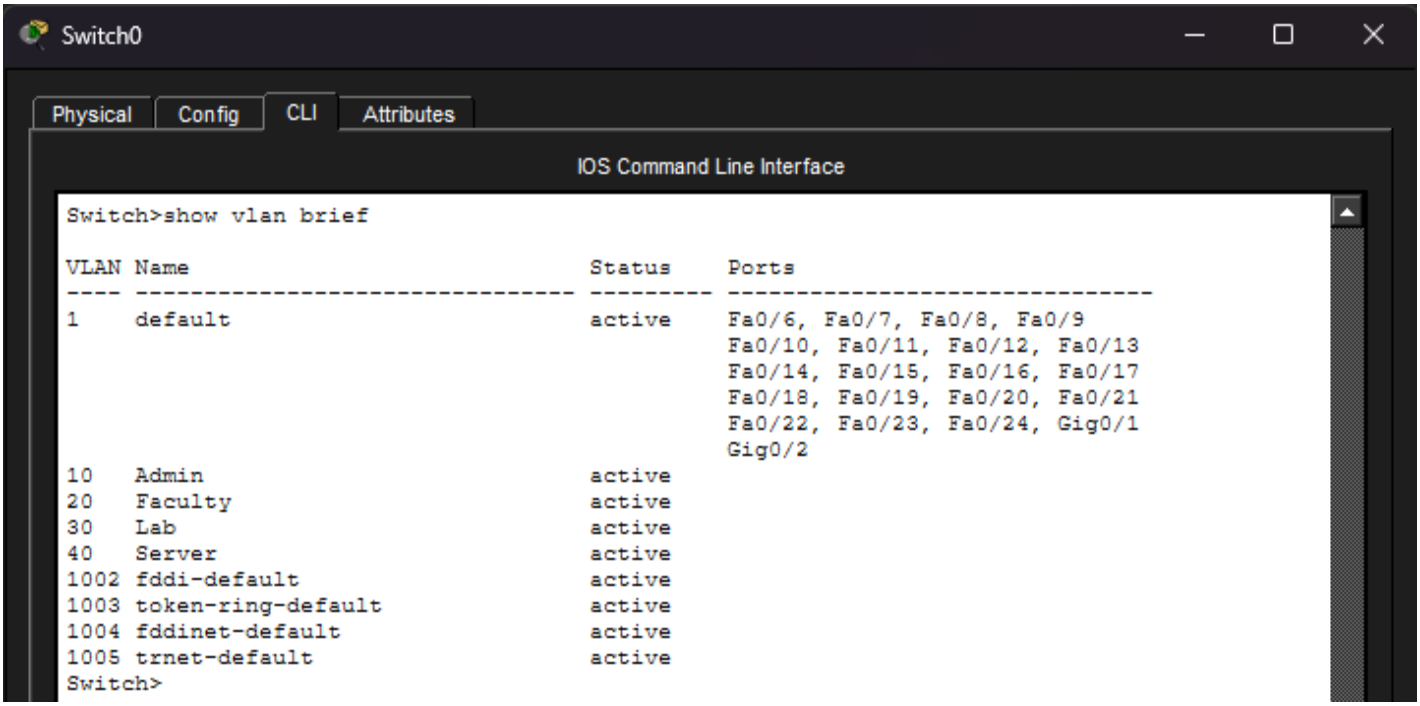

Router0
Physical
Config
CLI
Attributes
IOS Command Line Interface
Router(config)#interface gigabitEthernet 0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
exit
Router(config)#interface gigabitEthernet 0/1
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
exit
Router(config)#enable
% Incomplete command.
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#access-list 10 deny host 192.168.1.5
Router(config)#access-list 10 permit any
Router(config)#interface gigabitEthernet 0/1
Router(config-if)#ip access-group 10 out
Router(config-if)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
show access-lists
Standard IP access list 10
10 deny host 192.168.1.5
20 permit any
Router#
Copy
Paste
Top

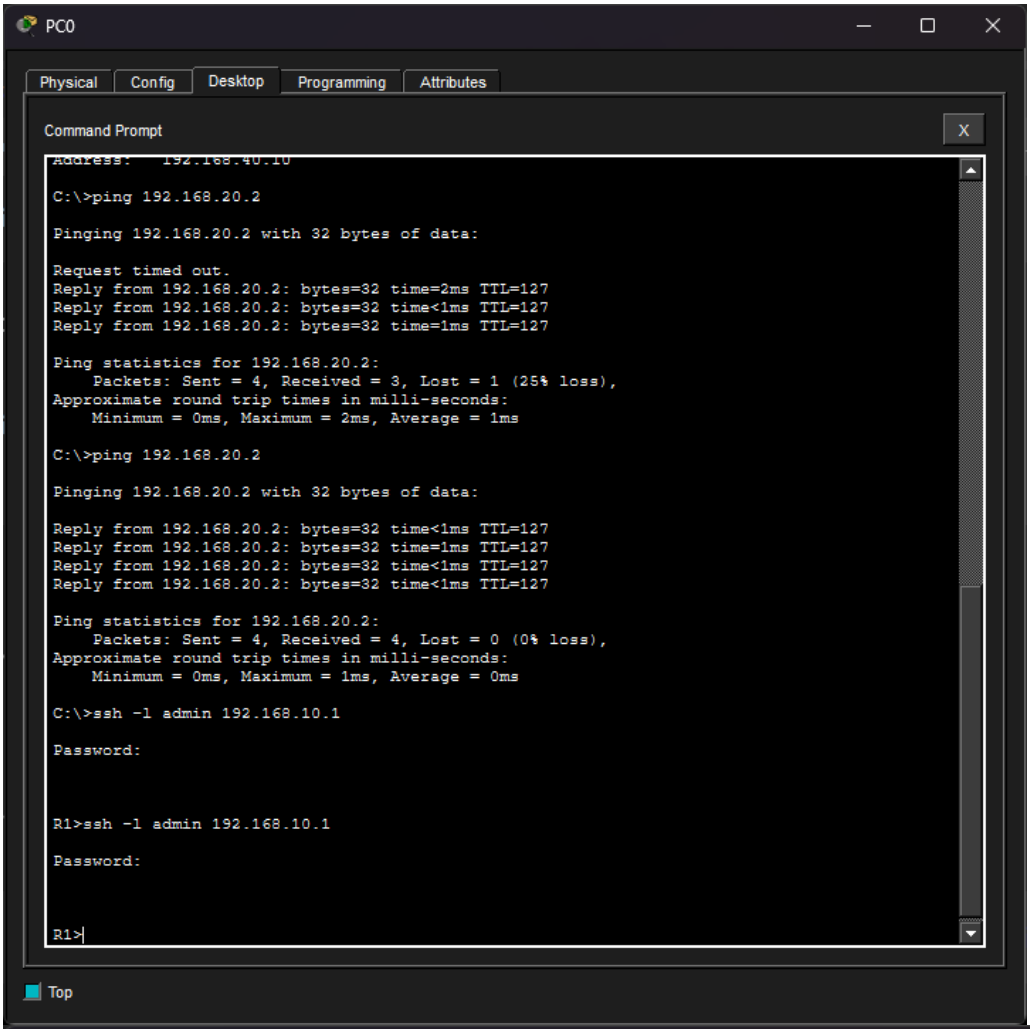

Router0
Physical
Config
CLI
Attributes
IOS Command Line Interface
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#access-list 10 deny host 192.168.1.5
Router(config)#access-list 10 permit any
Router(config)#interface gigabitEthernet 0/1
Router(config-if)#ip access-group 10 out
Router(config-if)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
show access-lists
Standard IP access list 10
10 deny host 192.168.1.5
20 permit any
Router#enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip access-list extended FIREWALL
Router(config-ext-nacl)#permit tcp 192.168.1.0 0.0.0.255 host 192.168.2.10 eq 80
Router(config-ext-nacl)#deny ip any any
Router(config-ext-nacl)#exit
Router(config)#interface gigabitEthernet 0/0
Router(config-if)#ip access-group FIREWALL in
Router(config-if)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
show access-lists
Standard IP access list 10
10 deny host 192.168.1.5 (4 match(es))
20 permit any
Extended IP access list FIREWALL
10 permit tcp 192.168.1.0 0.0.0.255 host 192.168.2.10 eq www
20 deny ip any any
Router#
Copy
Paste
Top

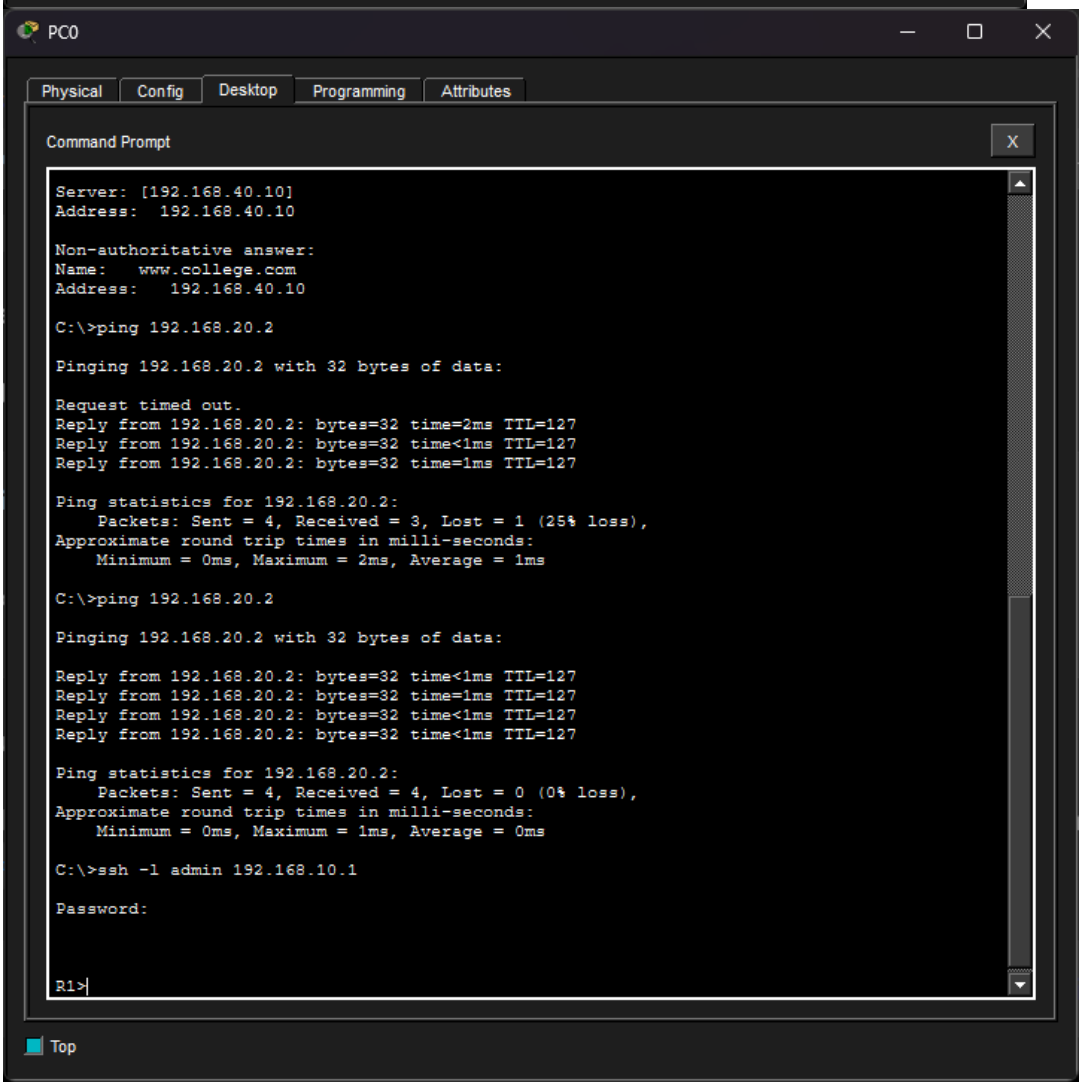

PC1
Physical
Config
Desktop
Programming
Attributes
Command Prompt
X
Reply from 192.168.2.10: bytes=32 time<1ms TTL=127
Reply from 192.168.2.10: bytes=32 time<1ms TTL=127
Reply from 192.168.2.10: bytes=32 time=1ms TTL=127
Ping statistics for 192.168.2.10:
Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>ping 192.168.2.10
Pinging 192.168.2.10 with 32 bytes of data:
Reply from 192.168.2.10: bytes=32 time<1ms TTL=127
Reply from 192.168.2.10: bytes=32 time=1ms TTL=127
Reply from 192.168.2.10: bytes=32 time<1ms TTL=127
Reply from 192.168.2.10: bytes=32 time=1ms TTL=127
Ping statistics for 192.168.2.10:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>ssh -l admin 192.168.1.1
% Connection timed out; remote host not responding
C:\>ssh -l admin 192.168.1.1
% Connection timed out; remote host not responding
C:\>ssh -l admin 192.168.1.1
% Connection timed out; remote host not responding
C:\>ssh -l admin 192.168.1.1
Password:
R1>telnet 192.168.1.1
Trying 192.168.1.1 ...Open
[Connection to 192.168.1.1 closed by foreign host]
R1>
Top

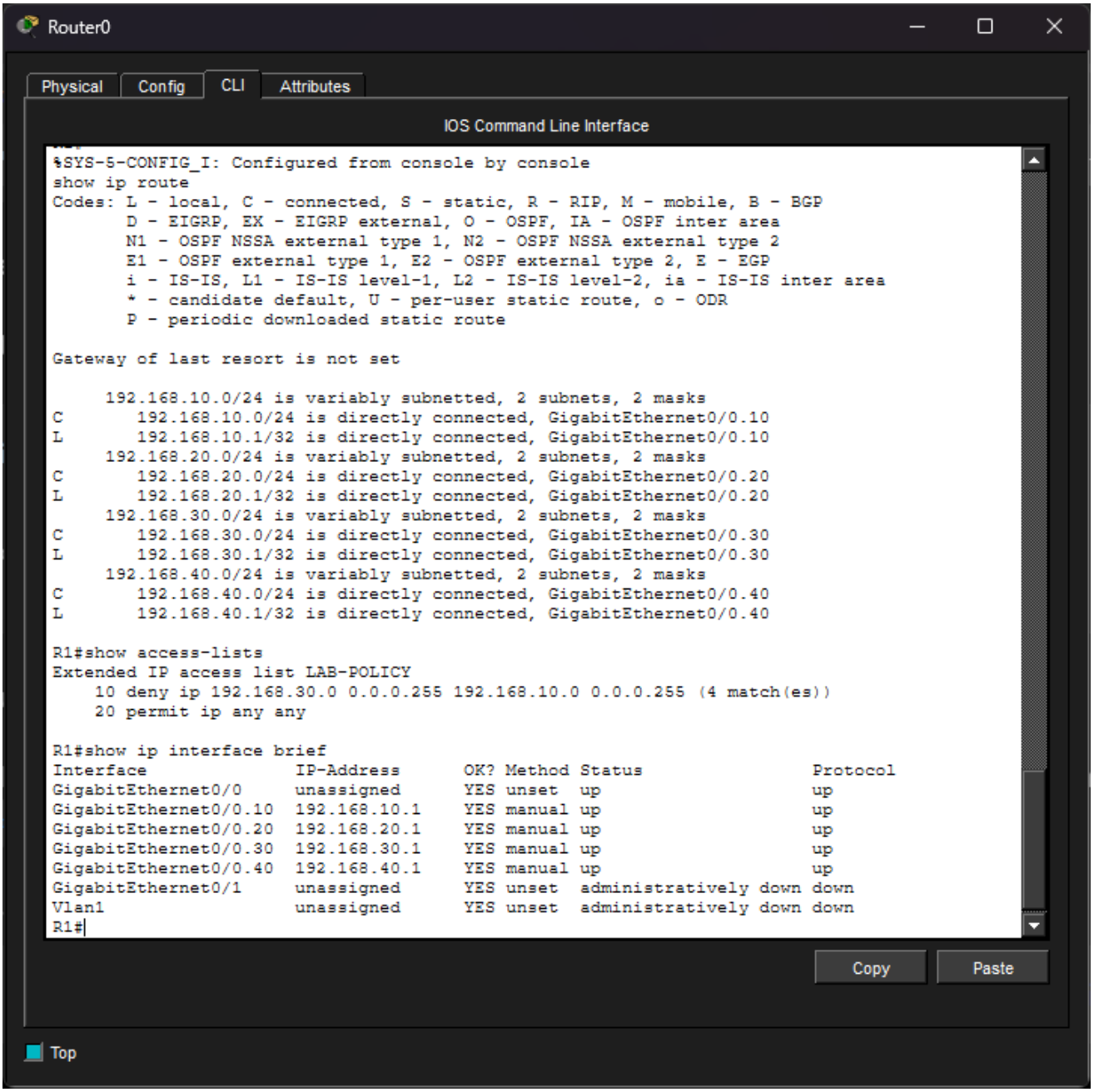
1. Complete topology






PC0
Physical
Config
Desktop
Programming
Attributes
Command Prompt
X
Address: 192.168.40.10
C:\>ping 192.168.20.2
Pinging 192.168.20.2 with 32 bytes of data:
Request timed out.
Reply from 192.168.20.2: bytes=32 time=2ms TTL=127
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127
Reply from 192.168.20.2: bytes=32 time=1ms TTL=127
Ping statistics for 192.168.20.2:
Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 2ms, Average = 1ms
C:\>ping 192.168.20.2
Pinging 192.168.20.2 with 32 bytes of data:
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127
Reply from 192.168.20.2: bytes=32 time=1ms TTL=127
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127
Ping statistics for 192.168.20.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>ssh -l admin 192.168.10.1
Password:
R1>ssh -l admin 192.168.10.1
Password:
R1>
Top


PC0
Physical
Config
Desktop
Programming
Attributes
Command Prompt
X
Server: [192.168.40.10]
Address: 192.168.40.10
Non-authoritative answer:
Name: www.college.com
Address: 192.168.40.10
C:\>ping 192.168.20.2
Pinging 192.168.20.2 with 32 bytes of data:
Request timed out.
Reply from 192.168.20.2: bytes=32 time=2ms TTL=127
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127
Reply from 192.168.20.2: bytes=32 time=1ms TTL=127
Ping statistics for 192.168.20.2:
Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 2ms, Average = 1ms
C:\>ping 192.168.20.2
Pinging 192.168.20.2 with 32 bytes of data:
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127
Reply from 192.168.20.2: bytes=32 time=1ms TTL=127
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127
Ping statistics for 192.168.20.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>ssh -l admin 192.168.10.1
Password:
R1>
Top


Router0
Physical
Config
CLI
Attributes
IOS Command Line Interface
%SYS-5-CONFIG_I: Configured from console by console
show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route
Gateway of last resort is not set
192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.10.0/24 is directly connected, GigabitEthernet0/0.10
L 192.168.10.1/32 is directly connected, GigabitEthernet0/0.10
192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.20.0/24 is directly connected, GigabitEthernet0/0.20
L 192.168.20.1/32 is directly connected, GigabitEthernet0/0.20
192.168.30.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.30.0/24 is directly connected, GigabitEthernet0/0.30
L 192.168.30.1/32 is directly connected, GigabitEthernet0/0.30
192.168.40.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.40.0/24 is directly connected, GigabitEthernet0/0.40
L 192.168.40.1/32 is directly connected, GigabitEthernet0/0.40
R1#show access-lists
Extended IP access list LAB-POLICY
10 deny ip 192.168.30.0 0.0.0.255 192.168.10.0 0.0.0.255 (4 match(es))
20 permit ip any any
R1#show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 unassigned YES unset up up
GigabitEthernet0/0.10 192.168.10.1 YES manual up up
GigabitEthernet0/0.20 192.168.20.1 YES manual up up
GigabitEthernet0/0.30 192.168.30.1 YES manual up up
GigabitEthernet0/0.40 192.168.40.1 YES manual up up
GigabitEthernet0/1 unassigned YES unset administratively down down
Vlan1 unassigned YES unset administratively down down
R1#
Copy
Paste
Top